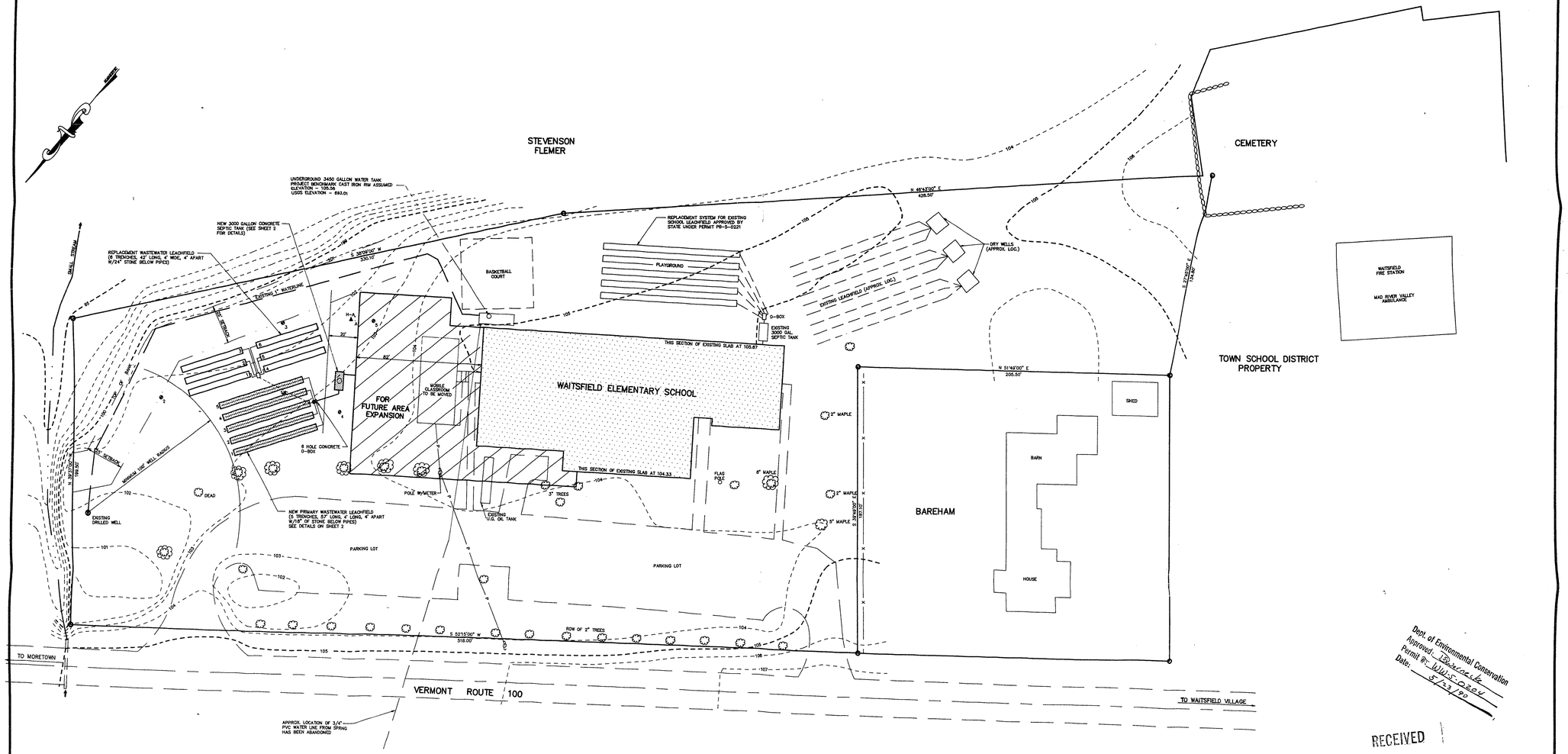
WW-5-0204



Dept of Environmental Conservation
Approved: [illegible]
Permit #: WWS-0224
Date: 5/23/90

RECEIVED
APR 18 1990
DIVISION OF PROTECTION

SITE PLAN
GRAPHIC SCALE

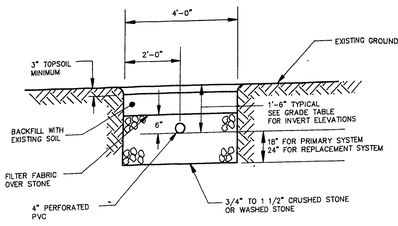
30 0 15 30 60 120
(IN FEET)
1 inch = 30 ft.

STATE OF VERMONT
CHARLES J GRENIER NO. 3248
REGISTERED ENGINEER
Charles J. Grenier

WATER AND SEWER SYSTEM LAYOUT
WAITSFIELD ELEMENTARY SCHOOL

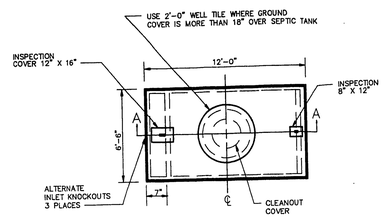
CHARLES GRENIER
CONSULTING ENGINEER
BOX 445, WATERBURY, VERMONT
244-6413

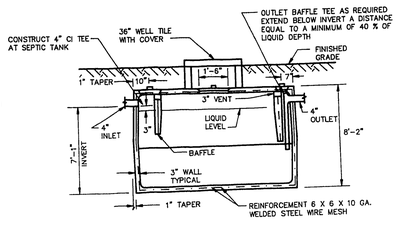
DATE = 4-17-90
DRN BY = T.M.
SCALE = 1" = 30'
SHT 1 OF 2



TYPICAL ABSORPTION TRENCH DETAIL

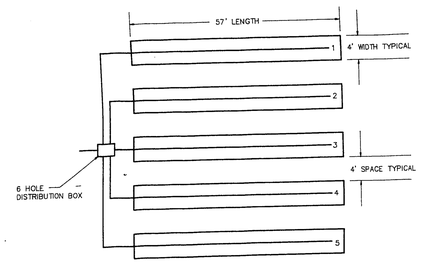
NTS



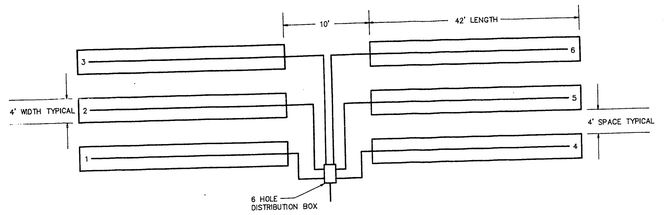


3000 GALLON SEPTIC TANK DETAIL

NTS



PRIMARY TYPICAL TRENCH LAYOUT DETAIL

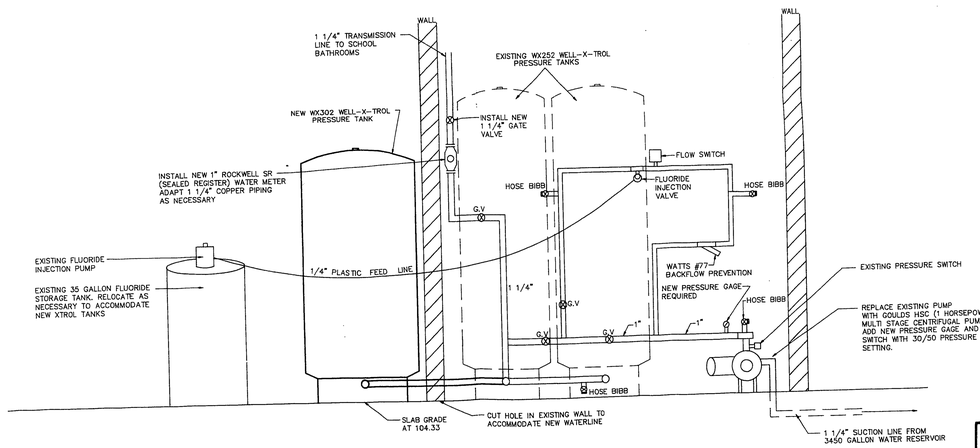


REPLACEMENT TYPICAL TRENCH LAYOUT DETAIL

NTS

GRADE TABLE

SYSTEM	COMPONENT		FLOWLINE GRADE
BUILDING ADDITION SLAB		=	105.67
PRIMARY	SEPTIC TANK INLET	=	102.90
	SEPTIC TANK OUTLET	=	102.65
	INTO D. BOX	=	102.40
	LINE 1	=	102.30
	LINE 2	=	102.30
	LINE 3	=	102.30
	LINE 4	=	102.30
	LINE 5	=	102.30
REPLACEMENT	INTO D. BOX	=	101.80
	LINE 1	=	101.60
	LINE 2	=	101.40
	LINE 3	=	101.20
	LINE 4	=	101.60
	LINE 5	=	101.40
	LINE 6	=	101.20



WATER SYSTEM SCHEMATIC

NTS

SEPTIC SYSTEM SPECIFICATIONS

1. ALL SEPTIC SYSTEM AREA TOPOGRAPHY SHOWN ON THIS SITE PLAN WAS DONE BY A FIELD CONTROL SURVEY PREPARED BY OUR OFFICE, BASED UPON A PROPERTY SURVEY BY JOHN ROTH OF WARREN, VERMONT. THE VERTICAL CONTROL FOR THIS PROJECT IS ASSUMED.

2. SEPTIC TANKS: ALL SEPTIC TANKS SHALL BE OF WATER TIGHT CONSTRUCTION AND MADE OF REINFORCED CONCRETE WITH A MINIMUM COMPRESSIVE STRENGTH OF 4,000 PSI AT 28 DAYS. ALL TANKS SHALL BE REINFORCED USING WIRE MESH OR REINFORCING STEEL AS NECESSARY TO ENSURE STRUCTURAL INTEGRITY. ALL SHIPLAP JOINTS SHALL BE SEALED WITH ASPHALT CEMENT OR EQUIVALENT. THE SEPTIC TANKS SHALL BE AS BUILT BY CAMP PRECAST OF MILTON, S.T. GRISWOLD OF WILLISTON, OR APPROVED EQUAL. INCOMING AND EXITING PIPE HOLES SHALL BE SEALED. SEPTIC TANK ACCESS COVERS AND DISTRIBUTION BOXES SHALL BE REINFORCED CONCRETE.

3. LEACHFIELD: DISTRIBUTION PIPING FOR LEACHFIELD SHALL BE 4" DIAMETER PERFORATED PVC SOIL AND DRAIN PIPE LAID TO A SLOPE OF 1/8" PER 10'. 4" SOLID PVC PIPE SHALL BE USED BETWEEN THE D BOX AND THE LEACHFIELD. A SHEET OF FILTER FABRIC SHALL BE PLACED OVER THE CRUSHED STONE BEFORE PLACEMENT OF ANY FILL. THE TRENCHES SHALL BE DUG TO THE PROPER GRADE SHOWN ON THE PLANS USING A BACKHOE. THE BOTTOM AND SIDE WALLS SHALL BE RAKED TO ENSURE ELIMINATION OF ANY BUCKET SMEARS. USING THE BUCKET ON THE BACKHOE OR FRONT END LOADER, DUMP THE STONE IN THE TRENCHES TO AN ELEVATION APPROXIMATELY 4" ABOVE THE FLOW LINE OF THE DISTRIBUTION PIPES. AT THE LOCATION OF THE DISTRIBUTION PIPE, MAKE CHANNELS ABOUT 4" DEEP BY USING A SHOVEL. LAY THE DISTRIBUTION PIPE IN THE TROUGH TAKING CARE TO LEVEL IT WITH HOLES DOWNWARD. REMOVE DIPS AND RISES WHICH OCCUR IN THE PIPE. FINISH PLACING THE CRUSHED STONE 2" DEEP OVER THE DISTRIBUTION PIPE. AFTER THE DISTRIBUTION SYSTEM HAS BEEN INSPECTED BY THE ENGINEER, PLACE 12" OF SOIL OVER THE TRENCHES, FOLLOWED BY AT LEAST 3" OF TOPSOIL OVER THE ENTIRE BED AREA. SHAPE THE LEACHFIELD SO THAT THERE WILL BE NO STANDING SURFACE WATER.

4. FINAL GRADING: ALL DISTURBED AREA AROUND THE LEACHFIELD SHALL BE FINE GRADED TO ENSURE CONTOURS. THIS AREA SHALL THEN BE SEEDED WITH A GRASS MIXTURE AND A LIGHT COVERING OF MULCH PLACED TO ENSURE THE GERMINATION OF SEEDS.

5. ALL SEWER CONSTRUCTION SHALL CONFORM TO EITHER 10 STATE STANDARDS FOR SEWAGE WORKS, 1978, OR ENVIRONMENTAL PROTECTION RULES. ALL WATER LINES SHALL BE INSTALLED AS PER 10 STATE STANDARDS FOR THE WATER WORKS, 1976, THE NATIONAL PLUMBING CODE, 1968 OR AWWA STANDARDS.

6. IT SHALL BE THE SCHOOLS RESPONSIBILITY TO CONFORM TO THE REQUIREMENTS OF THE PERMIT AND INSURE THAT THE PROJECT IS BUILT IN CONFORMANCE TO THE PLANS AND SPECIFICATIONS. THE ENGINEER SHALL RECIEVE 24 HOURS NOTICE PRIOR TO CONSTRUCTION OF KEY ELEMENTS OF THIS SEPTIC SYSTEM. CONDITIONS OF THE WATER SUPPLY AND WASTEWATER DISPOSAL PERMIT ARE TO BE FOLLOWED CAREFULLY. THE DEVELOPER SHALL SUBMIT PLANS TO THE WAITSFIELD HEALTH OFFICER FOR THE APPROVAL PRIOR TO COMMENCEMENT OF CONSTRUCTION.

BASIS OF DESIGN

1. BASIS OF DESIGN NEW SEPTIC SYSTEM ON NORTHSIDE OF SCHOOL
   A. DESIGN FLOW = 2250 GPD MAXIMUM (150 STUDENTS AND STAFF AT 15 GPD EACH)
   B. SEWAGE APPLICATION RATE = 1.5 GAL/SF/DAY
   C. PERCOLATION TEST = 2 MINUTES/INCH
   D. LEACH AREA REQUIRED = 1125 SF WITH 18" STONE, 990 SF WITH 24" STONE
   E. LEACH AREA PROVIDED = 1140 SF PRIMARY WITH 18" STONE, 1008 SF WITH 24" STONE
      PRIMARY SYSTEM (5 TRENCHES, 4' WIDE, 42' LONG)
      REPLACEMENT SYSTEM (6 TRENCHES, 4' WIDE, 42' LONG)
   F. SEPTIC TANK SIZE = 3000 GALLON CONCRETE
   G. PUMP STATION = NOT REQUIRED

2. BASIS OF DESIGN OF APPROVED REPLACEMENT SYSTEM ON SOUTH SIDE OF SCHOOL, APPROVED UNDER PERMIT #B-5-0221
   A. DESIGN FLOW = 2250 GPD MAXIMUM (150 STUDENTS AND STAFF AT 15 GPD EACH
   B. SEWAGE APPLICATION RATE = 1.25 GAL/SF/DAY
   C. PERCOLATION TEST = 4 MINUTES/INCH OR LESS
   D. LEACH AREA REQUIRED = 1800 SF
   E. LEACH AREA PROVIDED = 1800 SF (5 TRENCHES, 90' LONG, 4' WIDE)
   F. SEPTIC TANK SIZE = 3000 GALLON METAL (EXISTING)
   G. PUMP STATION = NOT REQUIRED

RECEIVED
APR 18 1990
Dept. of Environmental Conservation
Approved: B.W. Nemecke
Permit #: WW-5-0201
Date: 5/23/90
DIVISION OF PROTECTION

STATE OF VERMONT CHARLES J. GRENIER NO. 3248 REGISTERED ENGINEER

Charles J. Grenier

WATER AND SEWER SYSTEM DETAILS
WAITSFIELD ELEMENTARY SCHOOL

CHARLES GRENIER CONSULTING ENGINEER
BOX 445, WATERBURY, VERMONT
244-6413

DATE = 4-17-90
DRN BY = T. M.
SCALE = A. N.
SHT 2 OF 2

135275

WW-5-0204

PROJECT DOCKET SHEET

ID# WW-5-0204

DATE	REVIEW ACTION
4/18/90	Info rcvd
4/20/90	Logged in, Icopy to PS for PR sheet
5/23/90	Draft Permit

Project ID #: WW-5-0204 C = CO Applicant O = Owner S = Seller

APPLICANT

Name: Waitsfield School District

Address:

City, State, Zip: Waitsfield, VT 05673

CROSS REFERENCE

Town of Devt: WAITSFIELD Road/River: RT 100

Devt Name:

Description: school expansion

Fee Rec'd: $EXEMPT-MUN

DATES

Recd at Dist: 4/18/90
Logged/Rec Ltr: 4/20/90
Initial Review: 5 /23/90
Comp. Response: 5 /23/90
Final Action: 5 /24/90
Type of Final Action:
(See list of Final Action Codes Below)

Consulting Engineer: Charles Grenier; Box 445, Waterbury 05676

Hydrogeologist:

Reg. Review Eng: ~~EXEMPT~~ Wernecke

Action Requested: Engr

ACTION CODES

P = Permit
D = Denial
C = Certification of Compliance
N = Cert. of Noncompliance
W = Withdrawn

TO BE COMPLETED AFTER FINAL ACTION

Minor Project (Y/N) N
Type of Water Supply (ON/OFF/MUN/COM/NO): ON
Type of Sewage Disposal (ON/OFF/MUN/COM/NO): ON
Date Permit Recorded: _____

Number of Lots Approved: 0
Number of Mounds: 0
Gals. Sewage Generated: 1600

FOR SUBDIVISIONS ONLY:
List deferral number if this permits removes a deferral: _____

FOR DEFERRALS ONLY:
Is parcel retained or conveyed (R/C): _____
Is parcel contiguous or noncontiguous to other lands of buyer (C/N): _____

STREAM ALTERATION PERMITS ONLY:
Purpose: (GR=Gravel Removal, SM=Streambank Mod/Stab, CR=Channel Relocation, DI=Diversion, OT=Other) (GR/SM/CR/DI/OT): _____

State of Vermont

Department of Fish and Wildlife
Department of Forests, Parks and Recreation
Department of Environmental Conservation
State Geologist
Natural Resources Conservation Council

AGENCY OF NATURAL RESOURCES
Department of Environmental Conservation
Barre Regional Office
324 North Main Street
Barre, VT 05641

Telephone 479-3621

August 29, 1991

Charles Grenier
Box 445
Waterbury, VT 05676

Dear Charlie:

RE: WW-5-0204, Waitsfield Elementary School, Waitsfield

This letter is a follow up to the information submitted for the above-referenced permit.

Please be advised that condition 3.3 of the above referenced permit has been satisfied.

If you have any questions, please contact this office.

Sincerely,

Don Wernecke
Assistant Regional Engineer

CC Waitsfield School District

CHARLES GRENIER
CONSULTING ENGINEER PC
BOX 445 244-6413
WATERBURY, VT. 05676

August 13, 1991

RECEIVED
AUG 15 1991
DIVISION OF PROTECTION

Mr. Donald Wernecke, Assistant Regional Engineer
State of Vermont
Agency of Natural Resource
District #5 Environmental Office
324 North Main Street
Barre, Vermont 05641

RE: Waitsfield Elementary School Expansion, WW5-0204

Dear Don:

In accordance to the above mentioned permit, we staked out and inspected the installation of the wastewater disposal system for the School expansion. This leachfield consists of five trenches 57' long and 4' wide and is located on the north end of the School. It is our opinion that the system was installed in reasonable conformance to the approved plan. By writing this letter the Waitsfield Elementary School District is meeting Condition 3.2 involving the installation of the sewer system.

The water reservoir system tanks have been ordered and will be installed within the next week or so. We will complete the inspection of that system and submit a letter to you in the near future.

If any questions arise, please do not hesitate to contact us.

Respectfully,

Charles J. Grenier
Consulting Engineer, P.C.

CJG:lar

cc. Carol Hosford
John Rahill
Scotts Construction
Richard Kingsbury

State of Vermont

AGENCY OF NATURAL RESOURCES
WATER SUPPLY AND WASTEWATER DISPOSAL PERMIT

CASE No: WW-5-0204
APPLICANT: Waitsfield School District
ADDRESS: Waitsfield, VT 05673

LAWS/REGULATIONS INVOLVED
10 V.S.A. Chapter 61, Water Supply and Wastewater Disposal
Environmental Protection Rules
Chapter 4, Public Buildings
Chapter 7, Sewage Disposal
Chapter 8, Water Supply

This project, consisting of the construction of an addition to the Waitsfield Elementary School, located on Route 100, Waitsfield, Vermont, is hereby approved under the requirements of the regulations named above, subject to the following conditions:

1. GENERAL CONDITIONS

1.1. The project must be completed as shown on the plans prepared by Charles Grenier and listed as follows:
Water and Sewer System Layout dated 4/17/90,
Water and Sewer System Details, dated 4/17/90.
and which have been stamped "APPROVED" by the Division of Protection. No alteration of these plans shall be allowed except where written application has been made to the Agency of Environmental Conservation and approval obtained.

1.2. The school is approved for a maximum student population of 230 students. This limitation is based on the size of the water storage system. Should a water storage tank with a capacity of 4500 gallons or more be installed, the school could be approved for 300 students.

1.3. A copy of the approved plans and the Water Supply and Wastewater Disposal Permit shall remain on the project during all phases of construction and, upon request, shall be made available for inspection by State or Local personnel.

1.4. No alterations to the existing building other than those indicated on the approved plan, which would change or affect the interior waste plumbing, water supply, or wastewater disposal shall be allowed without prior review and approval from the Agency of Environmental Conservation.

1.5. This authorization does not relieve you, as applicant, from obtaining all approvals and permits as may be required from the Department of Labor and Industry (phone 828-2106), the Vermont Department Education, and local officials PRIOR to construction.

1.6. This permit shall in no way relieve you of the obligations of Title 10, Chapter 48, Subchapter 4, for the protection of groundwater.

2. WATER CONDITIONS

2.1. The lot is approved with the existing on-site water supply system provided the water is sampled and meets or exceeds potable water standards set forth by the Department of Health. All isolation distances as set forth in Environmental Protection Rules, Chapter 8, Section 8.08, shall be adhered to.

3. SEWAGE CONDITIONS

3.1. The project is approved for the on-site subsurface disposal of wastewater within the soil boring and percolation testing area indicated on the approved plans for a maximum of 4500 gallons of sewage per day. No buildings, roads, water lines, or other construction that might interfere with the installation or operation of the sewage disposal system are permitted. Isolation distances as set forth in Appendix 7-D of the Environmental Protection Rules shall be adhered to.

(CONTINUED)

3.2. The wastewater disposal system shall be constructed as shown on the approved plans and shall be operated at all times in a manner that will not permit the discharge of effluent onto the surface of the ground or into the waters of the State.

3.3. A Vermont registered Professional Engineer shall inspect the system and report in writing PRIOR to occupancy that all work was done according to the approved plans and this permit.

Timothy J. Burke, Commissioner
Department of Environmental Conservation

By _____ 5/24/90
Don Wernecke, Assistant Regional Engineer

CC Charles Grenier
VT Dept. of Labor & Industry
VT Dept. of Education
Town Planning Commission
Don Robisky

State of Vermont

479-3621
Agency of Natural Resources
Dept. of Environmental Conservation
324 North Main Street
Barre, Vermont 05641

Department of Fish and Wildlife
Department of Forests, Parks and Recreation
Department of Water Resources & Environmental Engineering
Natural Resources Conservation Council

April 20, 1990

Waitsfield School District
Waitsfield, VT 05673

Dear Applicant:

RE: WW-5-0204 , school expansion, WAITSFIELD

We received your completed application for the above referenced project on April 18, 1990, including a fee of $0. This application falls under the Water/Wastewater: Greater than 1000 GPD program and under the Performance Standards for this program area we have a maximum of 45 days of "in-house" time for our review of your application.

Based upon the present workload, we anticipate review of your project within the next two to three weeks.

If a Project Review Sheet is not attached to this letter, it will be sent under separate cover. A Project Review Sheet indicates other State agencies you should contact regarding any approvals you may need under their programs.

If you have not already done so, you should check with town officials regarding any local requirements you may need to meet.

DEPARTMENT OF ENVIRONMENTAL CONSERVATION

BY Alida J. Conti
Alida J. Conti, Administrative Secretary

CC Town Planning Commission
Charles Grenier

~~Fee paid by check #~~

Application #NW-S-0204
(office use only)

AGENCY OF NATURAL RESOURCES
Permits, Compliance & Protection Division

WATER SUPPLY AND WASTEWATER DISPOSAL

THERE IS TO BE NO SITE WORK OR CONSTRUCTION COMMENCED ON THIS PROJECT WITHOUT WRITTEN APPROVAL FROM THE AGENCY OF NATURAL RESOURCES.

(See instructions on back. Contact the District Environmental Office with questions.)

RECEIVED
APR 18 1990
DIVISION OF PROTECTION

LANDOWNER:
Name: Waitsfield School District

Address: Waitsfield, Vermont 05673
(Mailing)

Tele. No.: 496-3643

APPLICANT:
Name: Same

Address:
(Mailing)

Tele. No.:

Business Name (if other than above): N/A

CONSULTANT: (architect, engineer, plumber)
Name: Charles J. Grenier, Consulting Engineer, P.C. Tele. No. 244-6413

Address: P.O. Box 445, Waterbury, Vermont 05676
(mailing) (Street) (Town) (State) (Zip)

PROJECT:
Location: (Attach Map) Town: Waitsfield Road/Highway: Route 100

A. Size of Parcel (include all adjacent tracts of land which you may own, or control by lease option, etc.): 5.26± acres

B. When was the parcel created? 1960±

C. List any prior Environmental or Act 250 permits issued for A above:
Describe the project (be specific: see item #1 on back) **(Attach 2 sets of plans)**
PB5-0221, PB5-0221-1

TYPE OF WATER SOURCE: (municipal, offsite, community, onsite) onsite drilled well
(Attach approval - see #6 on back)

TYPE OF SEWAGE DISPOSAL: (municipal, offsite, community, onsite) onsite leachfield
(if municipal, attach approval - see #5 on back)

Number of gallons of sewage generated: 2250 GPD

PROJECT COST: (building and site improvements only. DO NOT include cost of land, plant equipment, etc.) **TOTAL COST:** $500,000±

APPLICATION FEE: (sewage flows: 0 to 499 gpd=$60.; 500 to 1000 gpd=$100.; over 1000 gpd=$175.) Make check payable to STATE OF VERMONT.
AMOUNT ENCLOSED $ N/A/Municipal project

April 12, 1990 Carol R. Holford
DATE **SIGNATURE OF LANDOWNER**

April 12, 1990 Carol R. Holford
DATE **SIGNATURE OF APPLICANT**

PROJECT INFORMATION SHEET
Water Supply and Wastewater Disposal

( ) 1. Description of Project (on application form):
Should include number of seats if a restaurant, or number of living units if an apartment complex, number of square feet, number of employees or similar figures, type of water supply, type of sewage disposal, description of process wastes. If project is a change in use, describe current use.

( ) 2. Location of Project on USGS map or Town Highway Map.

( ) 3. Plot plan (submit 2 copies) by a professional engineer registered in the Sate of Vermont showing:

- ( ) building location and lot dimensions;
- ( ) existing and proposed surface waters and drainage systems;
- ( ) location of existing and proposed sewage system and/or sewer lines;
- ( ) location of existing and proposed water system and water supply lines;
- ( ) five foot contours in the sewage disposal area;
- ( ) location of all soil testing and bedrock outcrops;
- ( ) use of adjoining lands and location of their water and sewage systems.

( ) 4. Interior plumbing and ventilation plans (submit 2 copies) by a registered architect or engineer or other qualified plumbing consultant:

- ( ) floor plans indicating use of each interior space and plumbing fixtures;
- ( ) wastewater and water supply layouts and isometrics or vertical riser diagrams;
- ( ) ventilation of (15 cfm/fixture in bathrooms);
- ( ) materials listing and fixture schedule;
- ( ) seating layout and kitchen layout.

( ) 5. Soil data and sewage system design information by a professional engineer registered in the State of Vermont (submit 2 copies):

- ( ) test borings and soils description, and percolation tests in the area of the proposed sewage system;
- ( ) design of the proposed sewage system based on maximum design flows and the soils information;
- ( ) design of existing sewage system and statement as to the capability and condition of the system;
- ( ) municipal approval for connection to sewage system;
- ( ) building sewer and lateral plan, profile, and sections, construction, material and testing specifications.

( ) 6. Water supply system designed by a registered professional engineer (submit 2 copies):

- ( ) approval by Health Department, Water Supply Section required. Contact the Health Department at 862-7220;
- ( ) a static level well or spring yield test;
- ( ) a water supply system design;
- ( ) report of ( ) chemical and/or ( ) bacteriological testing of water supply;
- ( ) design of existing water supply system and statement as to the capability and condition of the system;
- ( ) municipal approval for connection to water system.

( ) 7. All plans to be dated and signed by designer.

( ) 8. Application fee: Sewage flows: 0 to 499 gpd=$60.; 500 to 1000 gpd=$100.; over 1000 gpd=$175. Make check payable to the STATE OF VERMONT.

CHARLES GRENIER
CONSULTING ENGINEER PC
BOX 445 244-6413
WATERBURY, VT. 05676

April 17, 1990

RECEIVED
APR 18 1990
DIVISION OF PROTECTION

Don Wernecke, Assistant Regional Engineer
District #5 Environmental Office
324 North Main Street
Barre, Vermont 05641

RE: Waitsfield Elementary School Expansion

Dear Don:

As a result of today's meeting at the School, we have finalized the engineering design plans for the water and sewer system improvements necessary to accommodate the proposed expansion. To make your review easier, the following exhibits are offered.

1. A Water Supply and Wastewater Disposal permit application signed by School Principal, Carol Hosford. No application fee has been included because the School is municipally owned.

2. Water system design calculations dated 4/13/90 (two pages).

3. Soil and percolation test logs of January 25, 1983.

4. Cut sheets from the Goulds catalog for the existing booster pump at the School.

5. Cut sheets from the Goulds catalog for the proposed new booster pump.

6. Report from H.A. Manosh for School well drilled in 1987.

7. Engineering design plans prepared by our office consisting of two sheets.

In 1978 the Waitsfield Elementary School was expanded on the north end. It was concluded at that time that the existing septic system on the south end (Permit PB-5-0221) was adequate for 150 students and staff. A replacement septic system was approved at a location directly behind the School on the east side. We are proposing that an additional septic system be designed on the north

RECEIVED
APR 18 1990
DIVISION OF PROTECTION

end of the School to accommodate 150 students and staff with 2250 gallons per day design flow. The basis of design for both septic systems is outlined in notes on sheet 2 of the plans. New bathrooms for students and staff will be included with the proposed building renovation on the north end. Upon completion of new bathrooms, the wastewater flow will be split 50/50 between the north and south leachfield systems. Each classroom will be assigned a bathroom with the flow split evenly between the two systems. An additional factor of safety will be supplied by water saving fixtures in the new bathrooms on the north end. The septic system design is for a total of 300 students and staff. The architect has been assigned to enlarge the school at this time to approximately 200 students and staff at this time.

The water for the School is provided by a 10 GPM well drilled in 1987. A deep well pump delivers the water to an existing 3450 gallon water reservoir located easterly of the building. This reservoir, using the one day storage requirement, has the capacity to serve 230. To our knowledge, the water tank is in good condition. If and when the School is expanded to the 300 person limit, this reservoir should be replaced with a new 4500 gallon concrete water storage tank.

We are recommending a new Goulds centrifugal pump to meet the peak flow of 38 GPM at 30 PSI and one additional WX302 pressure tank. A new 1" water meter will be installed.

All things considered, the water and sewer system improvements for this expansion are relatively straight forward. Please process this application as soon as possible. If any questions arise as a result of this information, please do not hesitate to contact us.

Respectfully,

Charles J. Grenier
Consulting Engineer, P.C.

CJG:lar

cc. Carol Hosford
Richard Moser
Richard Burroughs

CHARLES GRENIER
Consulting Engineer
P. O. Box 445
WATERBURY, VERMONT 05676
Phone 244-6413

JOB WAITSFIELD ELEMENTARY SCHOOL
SHEET NO. 1 OF 2
CALCULATED BY CJG DATE 4-13-90
CHECKED BY _____ DATE _____
SCALE _____

WATER SYSTEM CAPACITY CHECK:

EXISTING SCHOOL:

BOYS BATHROOM:

2 WC	x 5	=	10 FU
2 LAV	x 2	=	4 FU
1 WALL URINAL	x 5	=	5 FU

GIRLS BATHROOM

3 WC	x 5	=	15 FU
2 LAV	x 2	=	4 FU
KITCHEN SINK		=	2 FU
JANITORS SINK		=	2 FU
SUBTOTAL		=	42 FU

RECEIVED
APR 18 1990
DIVISION OF PROTECTION

NEW SCHOOL ADDITION

BOYS BATHROOM

2 WC	x 5	=	10 FU
2 LAV	x 2	=	4 FU
WALL URINAL		=	5 FU

GIRLS BATHROOM

2 WC	x 5	=	10 FU
2 LAV	x 2	=	4 FU

TEACHERS BATHROOM

WC		=	5 FU
LAV		=	2 FU
SUBTOTAL		=	40 FU

TOTAL PROJECT = 82 FU
PEAK FLOW = 38 GPM

PRODUCED BY NCR INC., CANTON, MASS. 02021

CHARLES GRENIER
Consulting Engineer
P. O. Box 445
WATERBURY, VERMONT 05676
Phone 244-6413

JOB WARFIELD ELEMENTARY SCHOOL
SHEET NO. 2 OF 2
CALCULATED BY CJG DATE 4/13/90
CHECKED BY _____ DATE _____
SCALE _____

RECEIVED
APR 18 1990
DIVISION OF PROTECTIO

With 30/50 Pressure Switch
Existing Myers HC100 / Serial # 460-61 (1 HORSEPOWER)
Rated: 30 psi = 30 GPM
40 psi = 27.5 gpm
50 psi = 21 gpm
60 psi = 13 gpm
Recommend Replacing with new GOULDS HSC (1 Hp) Multi Stage Centrifugal
30 psi = 38 gpm (meets peak flow)
40 psi = 30 gpm
50 psi = 20 gpm

Pressure Tank Sizing: 2 - WX252 X-trols @ 30/50 = 26 gallons drawdown each
Add 1 new WX302 @ 30/50 = 26 gallons
3 TANKS X 26 = 78 gallons / 30 (average) = 2.6 minutes / run time

EXISTING 3450 gallon x'trol Reservoir is adequate for average daily design flow (230 staff + students times 15 gpd = 3450 gpd. Actual metered flows would probably find water use @ 50% ± or 2250 GPD.
ENCAPSULATE TO 4500 GALLON TANK WHEN POPULATION EXCEEDS 230
Deep Well Pump / Town 1987 (3 Hp Fairbanks / Morse @ 200')
Pump rate @ 50 psi with flow down at pump w/ actual 10 gpm well and 100' of water over pump, pump Rated @ 7.6 gpm
Deep Well Pump IS ADEQUATE

PRODUCT 3004 NCR Inc. Dublin, Ohio 43017

CHARLES GRENIER
CONSULTING ENGINEER
WATERBURY, VT. 05676 · 244-6413
BOX 445

RECEIVED
APR 18 1990
DIVISION OF PROTECTION

The following test pits were dug on January 25, 1983 at the Waitsfield Elementary School. All test pits were dug by a backhoe on the north end of the elementary school under the supervision of Charles Grenier and are located on the attached site plan.

TEST PIT #1

0-1' top soil
1-3' brown sandy soil
3-10' brown gravel
3½' drainage mottle
7½' maximum seasonal water table
9½' water
Perc Rate in gravel = 1 min/in.

TEST PIT #2

0-6" top soil
6"-1½' fine sand
1½-10' brown gravel
8' high water table
9½' water
Per Rate @ 2'2" = 2 min/in.

TEST PIT #3

0-1' top soil
1-2½' brown sand
2½-10'brown gravel
7' high seasonal water table
9' water
Perc Rate @ 3' = 1½ min/in.

TEST PIT #4

0-1' top soil
1-3' brown sand
3-10' brown gravel
7½' high water table
9½' water
Perc Rate in gravel = 1 min/in.

TEST PIT #5

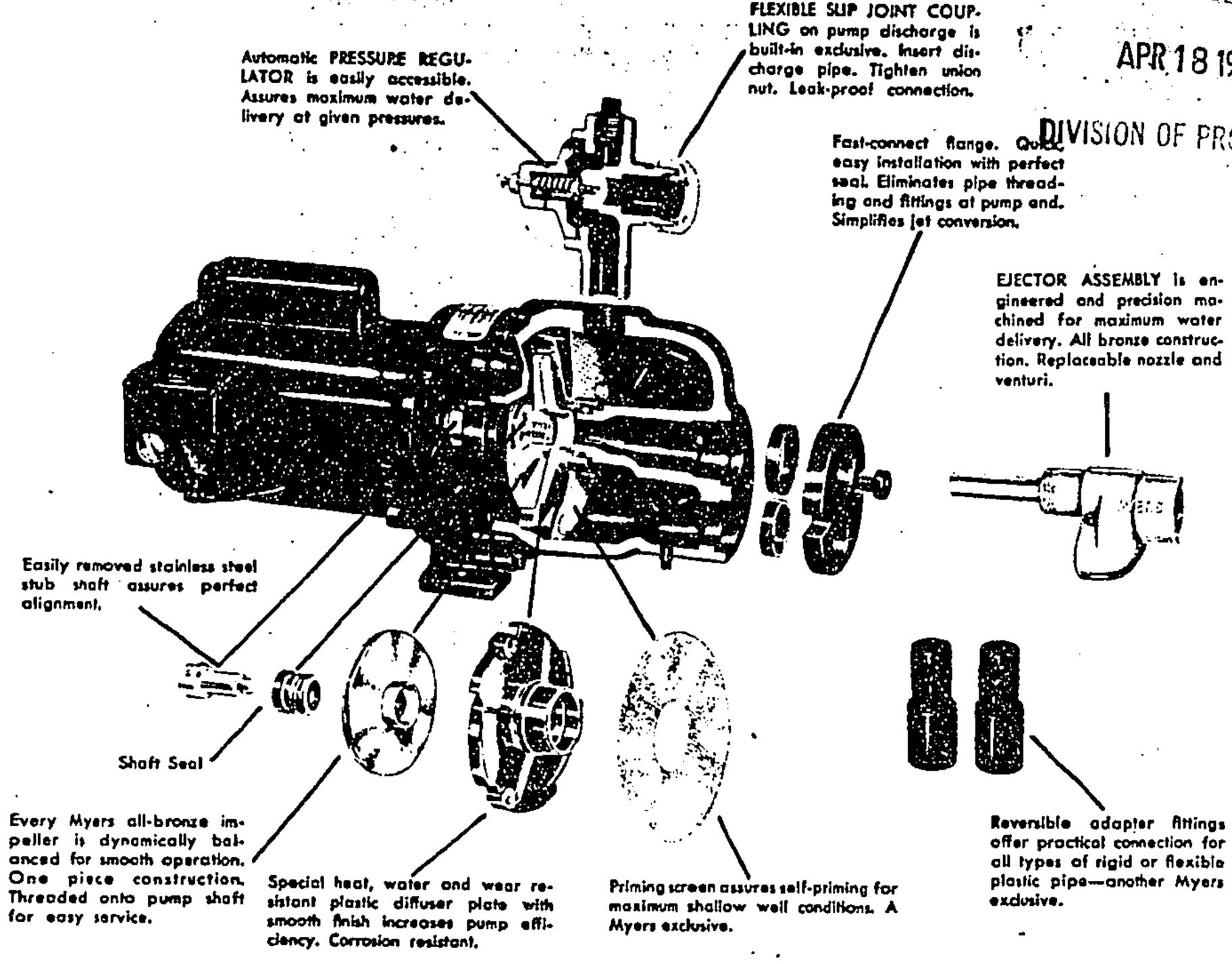
0-1' top soil
1-5' brown gray sand
5-9' brown gravel
9' water

102 | 4

WAITSFIELD ELEMENTARY SCHOOL - EXISTING PUMP

Myers HC Challenger Pumps

RECEIVED
APR 18 1990
DIVISION OF PROTECTION



The Myers Challenger paves the way for efficient and long-lasting water service. High quality materials are combined with a simplified construction to create outstanding performance on both shallow and deep wells. The Challenger is completely convertible. The same twin ejector assembly (the work horse of the jet pump) is used for both shallow and deep wells. A new improved impeller design means up to 25 per cent more water with no increase in motor load.

Check these other outstanding features:

√Ease of installation. The two Myers exclusives—**flexible slip joint coupling** and **reversible adapter fittings with fast-connect flange**—save installation time, materials and space (see above illustration).

√Self priming. Priming screen assures self-priming for maximum shallow well conditions. A Myers exclusive.

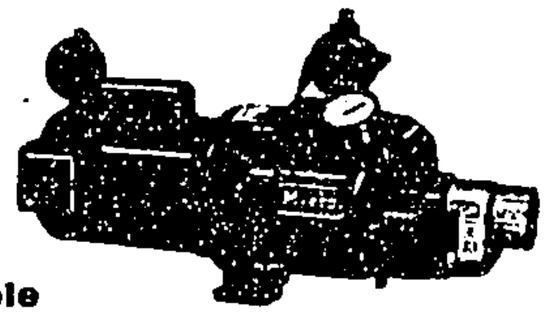
√Tailored Selection. Available in ½—½—¾ and 1 horsepower models. Capacities to 1800 gallons per hour.

The Myers Challenger offers value performance, advanced design.

And . . . it's built with the dependable perfection you expect from Myers.

"Fire-cured" Epoxy Coating

"Fire-cured" epoxy coating is used on all cast iron pump parts. This important Myers pioneered process was initially used to protect Myers sprayer tanks from highly corrosive spray chemicals. Now this same durable coating is a quality feature on all Myers Pumps.

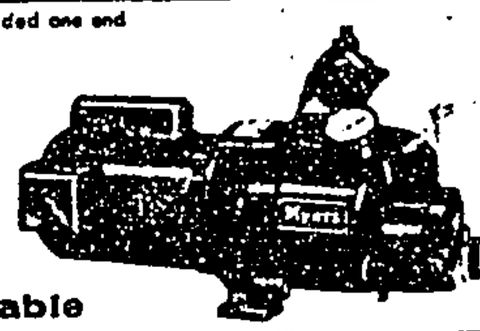


"HC" Shallow Well Selection Table

Basic Pump Catalog Number	Motor Horse Power	Jet Package Catalog Number	5 Foot Suction Lift Discharge Pressure, Lbs.						15 Foot Suction Lift Discharge Pressure, Lbs.						25 Foot Suction Lift Discharge Pressure, Lbs.					
		Bronze	20	30	40	50	60	70	20	30	40	50	60	70	20	30	40	50	60	70
			Capacity in U. S. Gallons Per Hour																	
HC33	⅓	SW4C3	780	780	535	260			610	610	415	160			375	375	280			
		A4C3	580	580	520	350	200		470	470	440	270	135		300	300	300	200		
HC50	½	SW4C5	1020	1020	840	420			810	810	630	240			480	480	450			
		A4C5	740	740	740	650	420	180	580	580	580	510	280		380	380	380	380	150	
		B4C5	380	380	380	380	380	320	300	300	300	300	300	280	200	200	200	200	200	200
HC75	¾	SW4C7	1500	1500	1350	900	330		1200	1200	1140	660			720	720	720	370		
		A4C7-10	1150	1150	1150	840	540	240	900	900	900	660	360		580	580	580	490	180	
		B4C7	660	660	660	660	540	410	530	530	530	530	500	360	340	340	340	340	340	300
HC100	1	SW4C10	1800	1800	1650	1250	780	135	1400	1400	1400	1080	540		870	870	870	870	300	
		A4C7-10	1150	1150	1150	1080	810	540	900	900	900	900	660	370	580	580	580	580	490	230
		B4C10	850	850	850	850	810	625	670	670	670	670	670	565	430	430	430	430	430	430

NOTE: A and B packages require pipe nipple when Ejector is mounted at pump. 1" by 2½" threaded one end for ⅓ and ½ horse power; 1¼" by 2½" threaded one end for ¾ and 1 horsepower.

RECEIVED
APR 18 1990
DIVISION OF PROTECTION



"HC" Deep Well Selection Table

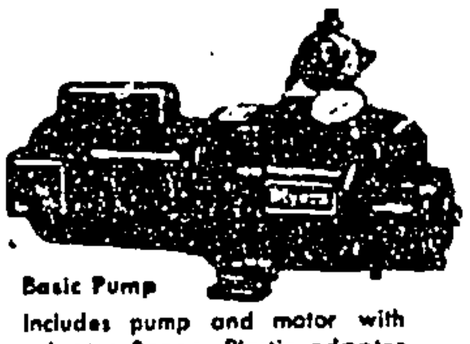
Basic Pump Catalog Number	Motor Horse Power	Jet Package Catalog No. Twin Type 4" Min. Well Bronze	Twin Type 4" Min. Well Cast Iron	Packer Type 3" Well	Packer Type 2½" Well	Packer Type 2" Well	Average Operating Pressure	Vertical Distance in Feet from Pump to Low Water Level 30	40	50	60	70	80	90	100	110	120
								Capacity in Gallons Per Hour									
HC33	⅓	A4C3	A4C3-I	JA3C3	JA25C3		20	630	540	440							
		B4C3	B4C3-I	JB3C3	JB25C3		20			390	345	280	220				
						JA2C3	20	600	480	380	290						
HC50	½	A4C5	A4C5-I	JA3C5	JA25C5		25	840	720	580							
		B4C5	B4C5-I	JB3C5	JB25C5		25			470	420	360	280				
						JA2C5	25	720	600	480							
						JB2C5	25			400	340	290	240				
HC75	¾	A4C7-10	A4C7-10-I	JA3C7-10	JA25C7-10		25	1100	960	780							
		B4C7	B4C7-I	JB3C7	JB25C7		25			740	625	520	420				
		C4C7	C4C7-I	JC3C7	JC25C7		30						400	340	270	200	
						JA2C7	30	800	630	510							
						JB2C7-10	35			480	390	320	250	200			
HC100	1	A4C7-10	A4C7-10-I	JA3C7-10	JA25C7-10		30	1240	1080	900							
		B4C-10	B4C-10-I	JB3C-10	JB25C-10		30			900	780	670	540				
		C4C-10	C4C-10-I	JC3C-10	JC25C-10		35						470	400	340	275	220
						JA2C-10	35	970	800	630							
						JB2C7-10	40			630	500	400	320	250	180		

"HC" Specifications

Basic Pump Catalog Number	Discharge	Pressure Switch Setting	Packer Type Inner Pipe Size For 2" Well	Packer Type Inner Pipe Size For 2½" & 3" Wells	Twin Type Drop Pipe Size	Foot Valve Size—Twin Type Only	Motor Capacitor Type—Built-in Overload Protection	Shpg. Wt. Pump Unit Only	Shipping Weight Jet Packages For 4" Well Bronze	4" Well C. I.	3" Well	2½" Well	2" Well
HC33	¾"	20-40	1"	1¼"	1¼" x 1"	1¼"	115 Volt	52	5	4	4½	4	4
HC50	¾"	20-40	1"	1¼"	1¼" x 1"	1¼"	115/230 Volt	54	5	4	4½	4	4
HC75	1"	20-40	1"	1¼"	1¼" x 1¼"	1¼"	115/230 Volt	68	5	4	4½	4	4
HC100	1"	20-40	1"	1¼"	1¼" x 1¼"	1¼"	115/230 Volt	71	5	4	4½	4	4

See Section 250 for Size and Catalog Numbers of Pressure Tanks.

NOTE: For Complete Water System—Order Basic Pump Unit, Proper Ejector Package, Pressure Tank and Air Control Separately.



Basic Pump

Includes pump and motor with adapter flange. Plastic adapter fittings. Pressure regulator valve. Pressure switch. Pressure gauge.



Twin Ejector Package

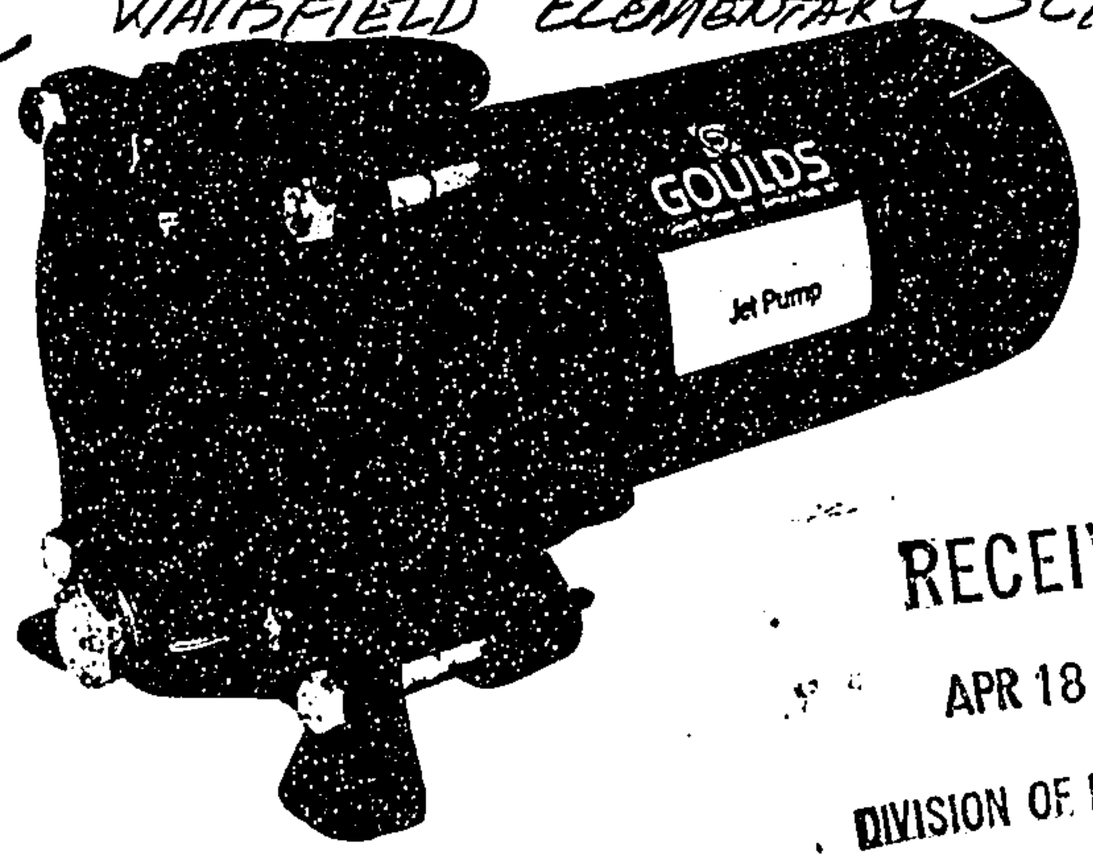
Includes all-bronze or cast iron ejector assembly and foot valve. Plastic nipple furnished in bronze package.

Bronze Packer Ejector

Includes all-bronze ejector assembly with built-in check valve and packer. For well adapters, see accessory section.



WAITSFIELD ELEMENTARY SCHOOL — NEW BOOSTER PUMP



Goulds

Multi-Stage Centrifugal Pumps

MODEL

HSC

RECEIVED
APR 18 1990
DIVISION OF PROTECTION

APPLICATIONS

Specifically designed for the following uses:

- Water circulation
- Booster service
- Liquid transfer
- Spraying systems
- Jockey pump service
- General purpose pumping

SPECIFICATIONS

Pump:

- Capacities to 50 GPM
- Pressures to 94 PSI (217 Feet)
- Pipe connections —
  1¼" suction
  1" discharge
- Temperatures: 180°F (82°C) max.
- Maximum working pressure: 125 PSI.
- Rotation: Right hand, ie; clockwise when viewed from motor end.

Motor:

- NEMA Standard
  ¾-2 HP, 115/230 Volt, 60 Hz
- Single Phase
- 3500 RPM
- Built-in Overload with Automatic Reset
- Capacitor Type
- Stainless Steel Shaft

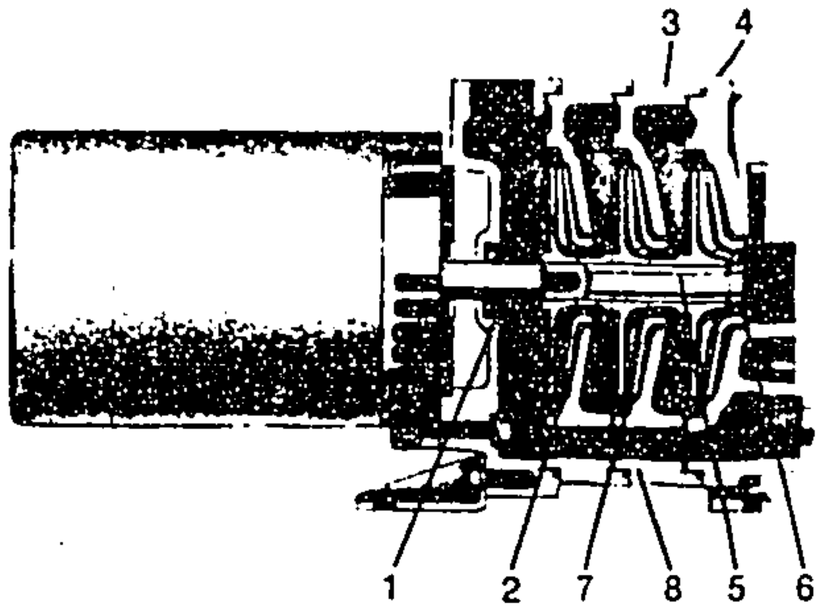
FEATURES

Multi-stage Design: Provides steady, quiet and vibration-free operation for years of trouble-free service.

Impellers: 20% glass-filled thermoplastic precision molded for high efficiencies.

Compact Design: Close-coupled, space saving design provides easy installation. Flexible coupling and bedplate not required.

Mounting: Can be mounted in vertical or horizontal position.



Stainless Steel Pump Shaft: Hex design provides positive drive for impellers and eliminates clearance adjustments.

Corrosion Resistant: Stainless steel wear rings and coverplates. Electro-coated paint process applied inside and out and then baked on.

O-Rings: Throughout for positive sealing.

Easy to Service: Can be taken apart for service by removing four bolts.

Motor: Close-coupled design. Ball bearings carry all radial/axial thrust loads. Designed for continuous operation. All ratings are within working limits of the motor.

HP	MODEL	STAGES
¾	HSC07	2
1	HSC10	2
1½	HSC15	2
2	HSC20	3

PARTS

1 Mechanical Seal
2 Corrosion Resistant Impellers
3 Rugged Cast Iron Construction
4 Electro-Coat Paint
5 Stainless Steel Hex Shaft
6 Stainless Steel Wear Rings
7 Stainless Steel Cover Plates
8 O-Ring Seals

©1988 Goulds Pumps, Inc.

Effective January, 1989

PERFORMANCE RATINGS

MODEL	TOTAL SUCTION LIFT (-FT.)	DISCHARGE PRESSURE — PSI 20	30	40	50	60	70	80	SHUT-OFF PRESS.
		CAPACITY — GALLONS PER HOUR							
HSC07	5	2400	1950	1380	420				52
	10	2280	1860	1260					50
	15	2190	1740	1080					48
	20	2100	1620	900					45
	25	2010	1500	600					43
HSC10	5	2640	2250	1800	1260	180			61
	10	2550	2160	1720	1140				59
	15	2460	2100	1620	950				57
	20	2380	1980	1500	720				54
	25	2300	1890	1020	480				52
HSC15	5	2880	2550	2190	1740	1140			69
	10	2790	2460	2100	1620	960			67
	15	2730	2400	1980	1500	780			65
	20	2670	2340	1920	1380	540			62
	25	2580	2220	1800	1200	180			60
HSC20	5	2940	2670	2430	2160	1860	1500	1050	92
	10	2880	2610	2370	2100	1800	1380	900	90
	15	2820	2550	2310	2010	1710	1290	780	88
	20	2760	2520	2250	1950	1620	1200	660	85
	25	2700	2460	2190	1890	1530	1080	480	83

CHOSEN * PUMP

Goulds
Multi-Stage Centrifugal Pumps

MODEL

HSC

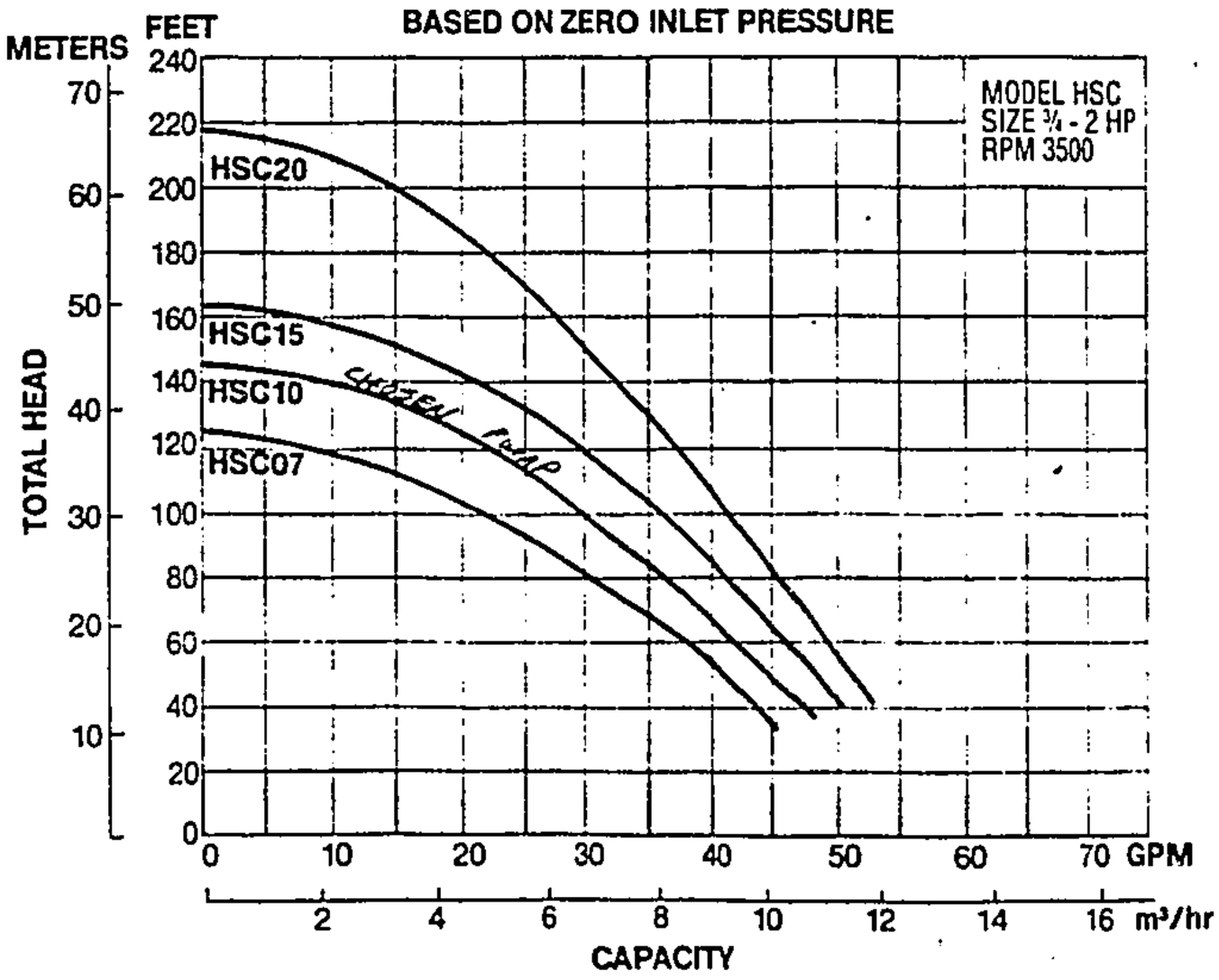
RECEIVED
APR 18 1990
DIVISION OF PROTECTION

DIMENSIONS AND WEIGHTS

HP/ Model	¾- HSC07	1- HSC10	1½- HSC15	2- HSC20
Weight	53	58	72	75
Length	15	16	17	19
Width	8	8	8	8
Height	9	9	9	9

(All dimensions in inches and weight in lbs.)
(Do not use for construction purposes)

PERFORMANCE CURVE



TO CONVERT FEET OF WATER TO PSI — DIVIDE FEET BY 2.31



PUMPS & ACCESSORIES
SEPTIC TANKS & PIPE

H.A. MANOSH, CORP.

RFD #1 Box 690
MORRISVILLE, VERMONT 05661
Telephone (802) 888-5722

WELL DRILLING • EXCAVATING

RECEIVED
APR 18 1990
DIVISION OF PROTECTION

Charlie Grenier, Eng. Box 445
Waterbury Vt.

Waitsfield Elem. School Well -

198' deep
97' of Casing
10 gpm.

Casing grouted into bedrock -, 10 feet in and
10 feet out -

AGENCY OF NATURAL RESOURCES AND ENVIRONMENTAL BOARD
PROJECT REVIEW SHEET

District# 5
Application # WW-5-0204

THIS IS NOT A PERMIT

Applicant Waitsfield School District
Town Waitsfield
Mail: Waitsfield, VT 05673

On 5/10/90 I reviewed information concerning a project on a tract/tracts of land of 5.26± acres, proposed by Waitsfield School District. The project will be on Lands owned by Same in Waitsfield, Vermont, and is generally described as:
Construct addition to elementary school. Onsite water supply + sewage disposal. On Rte 100.

Prior permits from this office: PB-5-0221 & amend.

A permit must be obtained for this project under the following permit programs which I have checked: (NO) ACT 250.
PERMITS/COMPLIANCE/PROTECTION PERMITS: _____ Deferral of Subdivision (yes) Water Supply & Wastewater Disposal _____ Subdivision _____ Tent/Trailer _____ Mobile Home Park _____ Stream Alteration _____ Municipal Sewer Line Ext. FISH & WILDLIFE PERMIT: _____ Stream Obstruction. THESE ARE THE ONLY PERMITS OBTAINABLE FROM THIS OFFICE. NO CONSTRUCTION OR SALE OF LOTS IS PERMITTED UNTIL ALL PERMITS CHECKED ABOVE HAVE BEEN ISSUED IN WRITING.

The above jurisdictional determination is based upon information provided and the following:

less than 10 ac(2) involved land. Env Bd Rule 2(A)(4)

THIS IS AN ADVISORY OPINION AND MAY BE APPEALED. ANY PARTY WHO DISAGREES WITH THIS OPINION MAY REQUEST A REVIEW DIRECTLY BY THE EXECUTIVE OFFICER OF THE ENVIRONMENTAL BOARD (ACT 250), OR THE DIRECTOR OF THE PROTECTION DIVISION, AS APPROPRIATE WITHIN 30 DAYS OF DECISION.

I also recommend that you contact the following Vermont State Agencies because this project may need approval under their programs or they may be able to assist you in preparing your project.

1. AGENCY OF NATURAL RESOURCES, Waterbury, VT 05676
   - _____ Air Pollution Control (244-8731)
   - _____ Hydro Electric Projects (244-7347)
   - _____ Dam Operations & Review (244-8755)
   - _____ Management of Lakes and Ponds (244-6951)
   - _____ Discharge of Temporary Pollution & Stormwater Permit (244-5674)
   - _____ Industrial Furnace/Boiler Conversion/ Installation (244-8731)
   - _____ Water Quality Certificate, if Federal permit req. (244-6951)
   - _____ Use of Chemicals in State Waters (244-5638)
   - _____ Bulk Fuel/Chemical Storage/Hazardous Wastes (244-8702)
   - _____ Industrial Process Air Emissions (244-8731)
   - _____ Indirect Discharge (greater than 6,500 gpd) (244-5674)
   - _____ Underground Storage Tanks (244-8702)
   - _____ Wetlands (244-6951)
   - _____ Waste Management Sanitary Landfills (244-8702)
2. DEPARTMENT OF LABOR & INDUSTRY, Montpelier, VT 05602 - Tel: 828-2106
   - ✓ Electrical Wiring Approval
   - ✓ Fire Prevention Approval
   - ✓ Access for Handicapped Approval
   - ✓ Plumbing in Public Buildings
   - ✓ Boilers and Pressure Vessels
   - _____ Storage of Flammable Liquids, Explosives
   - _____ Tramways and Ski Facilities
   - _____ Elevators
   - _____ Plumbing in single Family Residences Served by Public Water/Sewer w/ 10 or More customers
3. ✓ DEPARTMENT OF HEALTH, Burlington, VT 05401 - Tel: 863-7220 (Contact:: _____)
   - _____ Food, Lodging, Bakeries & Children's Camps
   - _____ Nursing Homes/Hospitals/Homes for Aged
   - _____ Protection of Public Water Source
   - _____ Extension of Water Lines over 500'
   - _____ Public Water systems
4. DISTRICT TRANSPORTATION ADMINISTRATOR (Consult the telephone directory under VERMONT, STATE OF, Highways)
   - _____ Access to State Highways (Residential Driveways)
   - _____ Development within 500' of a Limited Access Highway
5. AGENCY OF TRANSPORTATION, Montpelier, VT 05602 (Contact: _____)
   - _____ Signs (Travel Information Council) (828-2651)
   - _____ Airports and Landing Strips (828-2828)
   - _____ Junkyards (828-2587)
   - _____ Construction Within a State Highway Right-of-Way (Utilities, Signs, Driveways) (828-2653)
6. _____ DEPARTMENT OF AGRICULTURE, Montpelier, VT 05602 (Contact: _____)
   - _____ Use of Pesticides (828-2428)
   - _____ Slaughter Houses (828-2426)
   - _____ Primary Agricultural Soils (828-2504)
   - _____ Milk Processing Facilities (828-2433)
   - _____ Animal Shelters/Kennels (828-2421)
7. _____ AGENCY OF HUMAN SERVICES, Waterbury, VT 05676 Tel: 241-2158 (Contact: _____)
   - _____ Day Care Facilities
   - _____ Community Care Homes
   - _____ Residential Child Care Facilities
8. PUBLIC SERVICE BOARD (828-2358)/PUBLIC SERVICE DEPARTMENT (828-2811)
   - _____ Hydro Projects
   - _____ Alternative Energy Sources n
   - _____ Energy Conservation Measures
9. ✓ DIVISION FOR HISTORIC PRESERVATION, Montpelier, VT 05602 - Tel: 828-3226
10. ✓ LOCAL PERMITS - Contact Your Town Officials
11. OTHER: Dept. of Education

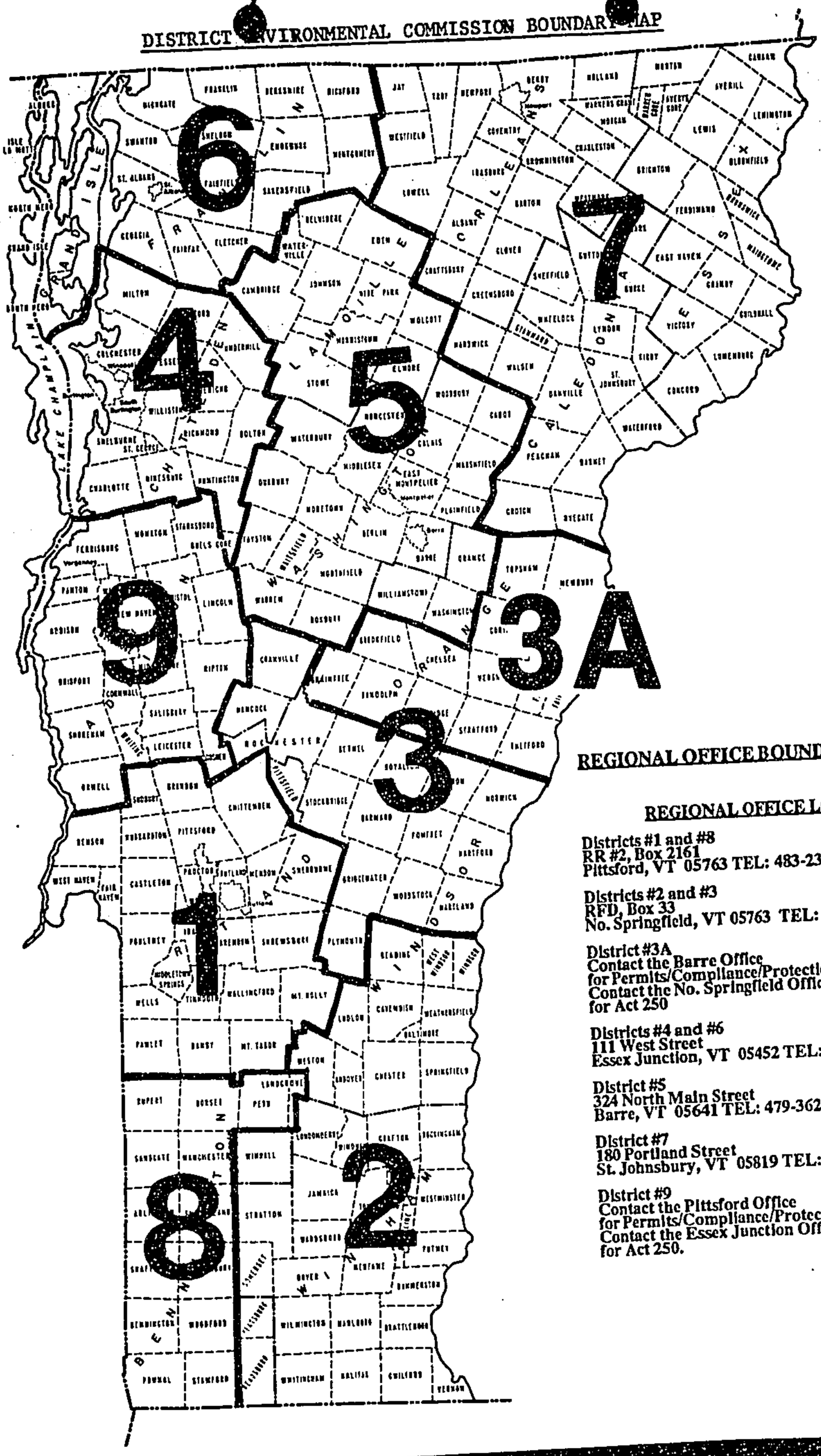
BY: Faye Ricke / Ms. Weiss
Protection Division/Environmental Board

Add: 324 No Main, Barre, VT 05641
Telephone Number 479-3621

Sent to Statutory Parties: YES _____ NO _____

3/89 White copy - Applicant Yellow copy - File Pink copy - Protection Division (Waterbury)

DISTRICT ENVIRONMENTAL COMMISSION BOUNDARY MAP



REGIONAL OFFICE BOUNDARY MAP

REGIONAL OFFICE LOCATIONS

Districts #1 and #8
RR #2, Box 2161
Pittsford, VT 05763 TEL: 483-2300

Districts #2 and #3
RFD, Box 33
No. Springfield, VT 05763 TEL: 886-2215

District #3A
Contact the Barre Office for Permits/Compliance/Protection
Contact the No. Springfield Office for Act 250

Districts #4 and #6
111 West Street
Essex Junction, VT 05452 TEL: 879-6563

District #5
324 North Main Street
Barre, VT 05641 TEL: 479-3621

District #7
180 Portland Street
St. Johnsbury, VT 05819 TEL: 748-8787

District #9
Contact the Pittsford Office for Permits/Compliance/Protection
Contact the Essex Junction Office for Act 250.